

**PES University, Bengaluru**

(Established under Karnataka Act 16 of 2013)

END SEMESTER ASSESSMENT (ESA) - May 2023**UE18CS353 - Object Oriented Analysis and Design with Software Engineering****Total Marks : 100.0**

1.a. Discuss different phases in the development of software. (5.0 Marks)

1.b. Illustrate the advantages and disadvantages of the Agile SCRUM model with an example. (5.0 Marks)

1.c. Differentiate the Incremental model and Evolutionary Model (5.0 Marks)

1.d. How will a software development project is planned using SE Management?
(5.0 Marks)

2.a. Consider a library management system. Draw a use case diagram that represents the primary functionalities of the system. Include at least three actors and five use cases. Provide appropriate labels and relationships between the actors and use cases. Briefly explain the significance of one of the relationships in the diagram.
(5.0 Marks)

2.b. Explain the various types of relationships that can be depicted between classes in a class diagram, and provide an illustrative example to support your explanation.
(5.0 Marks)

2.c. Consider a process for order fulfilment in an e-commerce system. Create an activity diagram that includes swimlanes, and demonstrates the use of fork and join constructs. Include at least three swimlanes representing different roles or departments involved in the process. Clearly depict the actions within each swimlane, and illustrate the parallelism and synchronization of activities using the fork and join constructs. Briefly explain the significance of using swimlanes, fork, and join in this activity diagram.
(5.0 Marks)

2.d. Elaborate on any one architectural design pattern with illustrative example
(5.0 Marks)

3.a. How does the Creator pattern in GRASP help in assigning responsibilities?
(5.0 Marks)

3.b. Describe the Liskov Substitution Principle (LSP) and its implications for object-oriented design.
(5.0 Marks)

3.c.

i. Match the following Design Principle names with their definition.

1. Information Expert	a. Combine related responsibilities into one manageable unit.
2. High Cohesion	b. Provide a well defined interface so that there will be no affect on other units
3. Indirection	c. Assign responsibility to the class which has the required data to fulfill that responsibility.
4. Protected Variation	d. Introduce an intermediate unit to communicate between the other units, so that the other units are not directly coupled.

ii. State True or False with justification:

- Protected Variation can be achieved using Polymorphism.
- Pure Fabrication classes cannot be reused in other systems.

(6.0 Marks)

3.d. Describe the role of version control in a Configuration Management System.
(4.0 Marks)

4.a. Describe the V-model of the software life cycle and discuss how testing activities are integrated into each stage.
(5.0 Marks)

4.b. Compare and contrast verification and validation in software testing. Provide examples to illustrate each concept.
(5.0 Marks)

4.c. Describe statement coverage and decision coverage as coverage-based test techniques. Discuss the differences between these two techniques and their importance in ensuring comprehensive test coverage. (5.0 Marks)

4.d. Explain the principles behind mutation testing and fault injection. Discuss their advantages and limitations as fault-based test techniques. (5.0 Marks)

5.a. Describe the ITIL service lifecycle and its five stages. Discuss the purpose and activities involved in each stage. (5.0 Marks)

5.b. Discuss the concept of affinity in DevOps and how it fosters cross-functional cooperation and shared responsibilities. Provide examples of how affinity can be established in a DevOps environment. (5.0 Marks)

5.c. Explain how continuous build, integration, and delivery contribute to shorter development cycles and faster time-to-market. Discuss the role of tools like Jenkins in facilitating these practices. (5.0 Marks)

5.d. Discuss the importance of automated build, testing, and deployment in the DevOps workflow. Explain how automation helps reduce errors and improve the reliability of software releases. (5.0 Marks)